TRELLIS: SENIOR ENGINEERING AUDIT & ACTION PLAN

Comprehensive Technical Review of BRD v1.0, SRS v1.0, System Architecture v0.1 & Prior Reviews

REVIEWER	TARGET STACK	DATE	VERDICT
Senior Principal Engineer	Go / Angular / PostgreSQL	October 23, 2026	Conditional Baseline

1. Executive Engineering Summary

The Trellis documentation suite exhibits exceptional domain discipline regarding user trust, safety invariants (manual publishing rules), and anti-hallucination controls. However, from a principal architecture standpoint, the project suffers from a **speculative baseline trap**: it freezes ~70 requirements before verifying fundamental third-party constraints (Instagram API scopes, free-tier LLM rate limits, and competitor data legality). This assessment provides an item-by-item audit of each document and prescribes precise architectural modifications for every gap identified.

> **Core Senior Engineering Directives**
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> **1. Uncouple Baseline from Unverified APIs:** Convert all third-party dependent requirements (Instagram, Growy, Free LLMs) into explicit contingent specifications backed by automated spikes.
> **2. Replace Qualitative Rules with Measurable Thresholds:** Eliminate subjective acceptance criteria in favor of bounded numerical SLAs, deterministic verification formulas, and structured fail-safes.

2. Business Requirements Document (BRD v1.0) Audit

SECTION / ITEM	STATUS	CURRENT DEFECT / RISK	SENIOR ENGINEERING ACTION PLAN
§1 Purpose & Roles	ADEQUATE	"Scrum Master" assigned to a solo build adds process overhead without delivery governance.	Reframe role as **Technical Lead / Developer**. Define single-developer Kanban cadences rather than empty Scrum ceremonies.
§2 Business Objectives	WEAK	Objectives lack quantitative metrics (e.g., "Reduce time and effort" has no baseline or goal).	Define baseline: "Reduce weekly planning duration from 180 mins to <20 mins on iPad." Measure time-to-first-draft and click depth.
§3 Stakeholder Scope	WEAK	In-scope competitor analysis relies on Growy without confirming access legality or pricing.	Make competitor scope conditional on **Spike B**. Add a fallback state: manual competitor handles entry with public web parsing.
§3 Form Factor Scope	WEAK	"Equivalent core functions" across iPad, desktop, and mobile doubles UI engineering scope.	Declare **iPad Safari as Primary Tier-1**. Mobile and Desktop are designated Tier-2 (functional, responsive layout, non-optimized).
BR-01 Profile Audit	ADEQUATE	"Actionable opportunities" is vague; lacks standard metrics taxonomy.	Define explicit Audit Data Contracts (profile bio, engagement rate, post frequency) using normalized Go structs.
BR-02 Comparisons	STRONG	Speculation prohibited as fact. Excellent trust boundary.	Maintain as-is. Enforce in backend by separating observed measurements from LLM inference tags in schema.
BR-06 Banned Words	STRONG	Hard prohibited phrase list governed strictly by the user.	Implement as an immutable server-side filter running prior to returning any LLM payload to the frontend.
BR-09 Competitor Rule	CRITICAL	Mandates latest 5 posts MUST be skincare. Unrealistic; 1 non-skincare post breaks evaluation.	**Modify Rule:** Require "at least 70% of recent 10 posts" to match skincare vector embeddings; ignore pinned/brand posts.
BR-10 Swiper Feature	WEAK	Bundles 6 distinct capabilities into one BR; lacks underlying data feed definition.	Decompose BR-10 into 3 distinct atomic user stories: (1) Data Ingestion, (2) Idea Scoring Engine, (3) Swiper UI Component.
BR-14 Usability	ADEQUATE	iPad touch targets and landscape/portrait requirements	Specify minimum touch target size of 44×44pt and enforce responsive breakpoints at 768px and

3. Software Requirements Specification (SRS v1.0) Audit

The SRS translates business goals into functional behaviors. While domain safety controls are robust, the spec frequently conflates product behavior with unverified technical constraints, particularly regarding free-tier AI mechanics and storage guarantees.

ID / MODULE	STATUS	REQUIREMENT DEFECT	SENIOR ENGINEERING MODIFICATION
AI-01 Intent	ADEQUATE	"Disclose consequential assumptions" lacks programmatically defined thresholds.	Define "consequential": any assumption altering metric calculations by >5% or introducing unverified product claims must carry a UI warning badge.
AI-04 Grounding	STRONG	Mandates deterministic computation for numbers. Core architectural rule.	Enforce via backend tool-calling: the LLM is prohibited from calculating sums/averages; Go executes mathematical primitives.
AI-06 AI Fallback	WEAK	"Bounded compatible fallback" under free provider limits is unengineered.	Implement an explicit circuit breaker: Tier 1 (Primary Free Model) → Timeout 8s → Tier 2 (Secondary Model) → Fallback to static template.
CONT-01 Package	ADEQUATE	Product-to-image association has no ground truth or error boundary.	Add explicit confidence score returned by vision model. If confidence <0.75, prompt user to confirm product identity via dropdown.
CONT-02 Experiences	STRONG	Replaces invented reviews with fillable placeholders `[User Experience]`.	Maintain as-is. Standardize template regex parsing in frontend editor to highlight unresolved placeholders in amber.
CONT-04 Voice	WEAK	"Learn writing patterns" implies model fine-tuning or complex training pipelines.	Rephrase: "Retrieve 5 most relevant past captions via PGvector cosine similarity and inject as Few-Shot examples into prompt context."
DATA-03 History	ADEQUATE	Relevance weighting algorithm for historical posts is unspecified.	Define decay function: $W = \text{similarity} \times e^{-\lambda t}$, balancing semantic relevance against temporal freshness.
DATA-06 Refresh	CRITICAL	Session-open analysis refresh breaks scheduled historical snapshot consistency.	**Decouple Data Fetching:** Background Go cron job fetches raw metrics daily (within API limits); session-open only triggers LLM narrative synthesis.
COMP-01/04	WEAK	Relies on third-party competitor endpoint that may violate Meta TOS.	Isolate competitor data layer behind `CompetitorProvider` interface. Implement mock provider immediately for local testing.
PLAN-03/04	STRONG	Zero autonomous publishing. Explicit manual action required per post version.	Store SHA-256 hash of published draft version. Publish API call fails if draft hash does not match user-approved hash.
PLAN-06 Autosave	ADEQUATE	Autosave debounce and network conflict handling unstated.	Implement 2000ms debounce on editor changes. Use optimistic locking with `version` column in PostgreSQL to reject stale writes.
NOTIF-01 Toast	ADEQUATE	Toast auto-dismissal risks missing actionable reminders while away.	Toasts auto-dismiss, but unresolved items persist in an unread "Action Center" badge in the main navigation bar.
AUTH-06 Retention	WEAK	Disconnect data retention choices lack specific database deletion queries.	Define precise cascade rules: Disconnect → Soft delete access tokens → Prompt retention choice → Hard delete cached posts if requested.
PERF-01 Latency	WEAK	"Should feel responsive" is untestable non-functional requirement.	Set quantitative SLA: Non-AI navigation <100ms (p95); Draft autosave <300ms (p95); AI Chat response stream start <1500ms.
COST-01	CRITICAL	Strict free-tier constraint without fallback	Implement token bucket rate limiter per user.

4. System Architecture (v0.1) Audit & Technical Modifications

The proposed architecture correctly selects a Go modular monolith and PostgreSQL. However, several critical operational mechanisms—specifically job queues, vector storage, media management, and resilience patterns—require immediate hardening.

ARCHITECTURAL ITEM	STATUS	STRUCTURAL RISK / GAP	SENIOR ENGINEERING TARGET STATE
ADR-001 Monolith	STRONG	Modular Go monolith is the correct architecture for this phase.	Enforce boundary integrity using Go internal packages (`internal/app/domain/...`) to prevent cross-module coupling.
ADR-002 Database	ADEQUATE	PostgreSQL standard store. Missing AI vector search capability.	Enable **pgvector** extension in PostgreSQL. Store caption/product embeddings directly in DB, avoiding pinecone/external vector DB.
ADR-003 Jobs	WEAK	PG-backed job table without locking mechanism causes duplicate execution.	Implement **SELECT FOR UPDATE SKIP LOCKED** worker queue pattern in Go to guarantee single-execution job semantics.
ADR-004 Adapters	STRONG	Provider-neutral interfaces for Instagram, AI, and News.	Define strict Go interfaces with mock implementations generated via `mockery` for 100% offline unit testing.
ADR-006 Publishing	STRONG	Immutable draft versioning before outbound Meta API call.	Enforce two-phase commit state machine: `Draft` → `Publishing_Intent` → `Provider_Ack` → `Published` / `Unknown_State`.
Media Storage	WEAK	Private media store vendor deferred; local disk vs S3 unclear.	Adopt S3-compatible Blob storage interface (`minio-go`). Use local filesystem in dev, Cloudflare R2 / AWS S3 in production.
Auth Architecture	CRITICAL	Auth implementation unselected; token encryption omitted.	Use Go native JWT session cookies with **AES-256-GCM encryption** for stored Instagram Long-Lived Access Tokens at rest.
LLM Orchestration	WEAK	Direct model calls without schema validation or rate limiting.	Implement an **AI Gateway Module** handling JSON-schema output enforcement, token counting, and prompt context trimming.
Error Handling	ADEQUATE	Ambiguous publish responses leave user uncertain.	In state `Unknown_State`, system calls GET `/media_id` on Meta API before permitting any manual retry attempt (Figure 3 expansion).
Deployment	WEAK	Deployment target unselected; sleeping hosts threaten scheduled tasks.	Target Single-Container Docker deployment on VPS (e.g., Hetzner/Fly.io) with persistent volume. Prevents container sleep issues.

5. Critique of Prior Claude Review (Meta-Review)

Evaluation of Claude's Review Findings

Accurate Assessment Points: Claude correctly flagged that the documents freeze requirements prematurely before validating third-party feasibility, rightly criticized the rigid "5/5 skincare posts" competitor rule, and identified unmeasured business goals.

Engineering Misalignments in Claude's Review: Claude recommended adding complex persona journeys and expanding multi-user onboarding. In a solo-developer build for an initial PO user, this introduces unnecessary complexity. The correct engineering response is to maintain laser focus on core functionality, deferring multi-tenant onboarding until product-market fit is proven.

6. Execution Roadmap & Technical Verification Spikes

To de-risk the project before committing to production feature development, the technical lead must execute four mandatory spikes. Development will proceed in three controlled phases governed by strict entry criteria.

Phase 0: Feasibility Spikes (Immediate - 2 Weeks)

SPIKE ID	TARGET DOMAIN	PASS/FAIL CRITERIA	FALLBACK PLAN IF FAILED
Spike A	Meta Graph API Scope	Successfully fetch profile insights, historical metrics, and trigger containerized image publishing on test account.	Pivot to manual analytics CSV upload; restrict app to content drafting and local planning.
Spike B	Competitor Feasibility	Fetch 10 recent public posts from target competitor handles without violating Meta platform TOS.	Remove automated competitor feed; allow user to manually paste competitor captions for analysis.
Spike C	Free LLM Benchmarking	Execute Vision-to-Caption and Analytics Tool-Calling within <5s latency and 0% rate limit lockouts over 50 test runs.	Introduce hybrid local model (Ollama/Llama-3) or configure user-provided OpenAI API Key (BYOK model).
Spike D	iPad Safari Prototyping	Verify 60fps touch swiping, local storage recovery, and zero layout shifting across portrait/landscape mode on iPad.	Replace custom touch gestures with explicit button-based controls (Keep / Pass buttons).

Phase 1 & 2 Execution Plan

PHASE	CORE DELIVERABLES	ENGINEERING GATES
Phase 1: Foundation (Weeks 3-6)	Go monolith skeleton, PG DB migrations, JWT Auth, Instagram OAuth linkage, Dashboard basic layout, Media upload pipeline.	100% unit test coverage on Auth/Domain logic; successful Instagram token refresh cycle.
Phase 2: Core Intelligence (Weeks 7-10)	AI Orchestrator module, pgvector integration, Caption Generator with banned-words filter, Manual Publishing state machine.	Zero banned words leaked in 100 test runs; publication state machine handles simulated network drops cleanly.
Phase 3: Polish & Beta (Weeks 11-12)	Swiper UI, Notification Action Center, iPad optimization, offline recovery, full verification test suite execution.	Pass all SRS scenarios (T-01 to T-10); PO sign-off on manual publishing workflow on physical iPad.

> **Final Senior Engineering Sign-off Thresholds**
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> The Trellis baseline specifications are hereby approved for development **ONLY AFTER** Spike A and Spike C report successful pass metrics. All technical debt and open decisions documented in this audit must be registered as tracked backlog items in the project repository prior to Sprint 1 start.